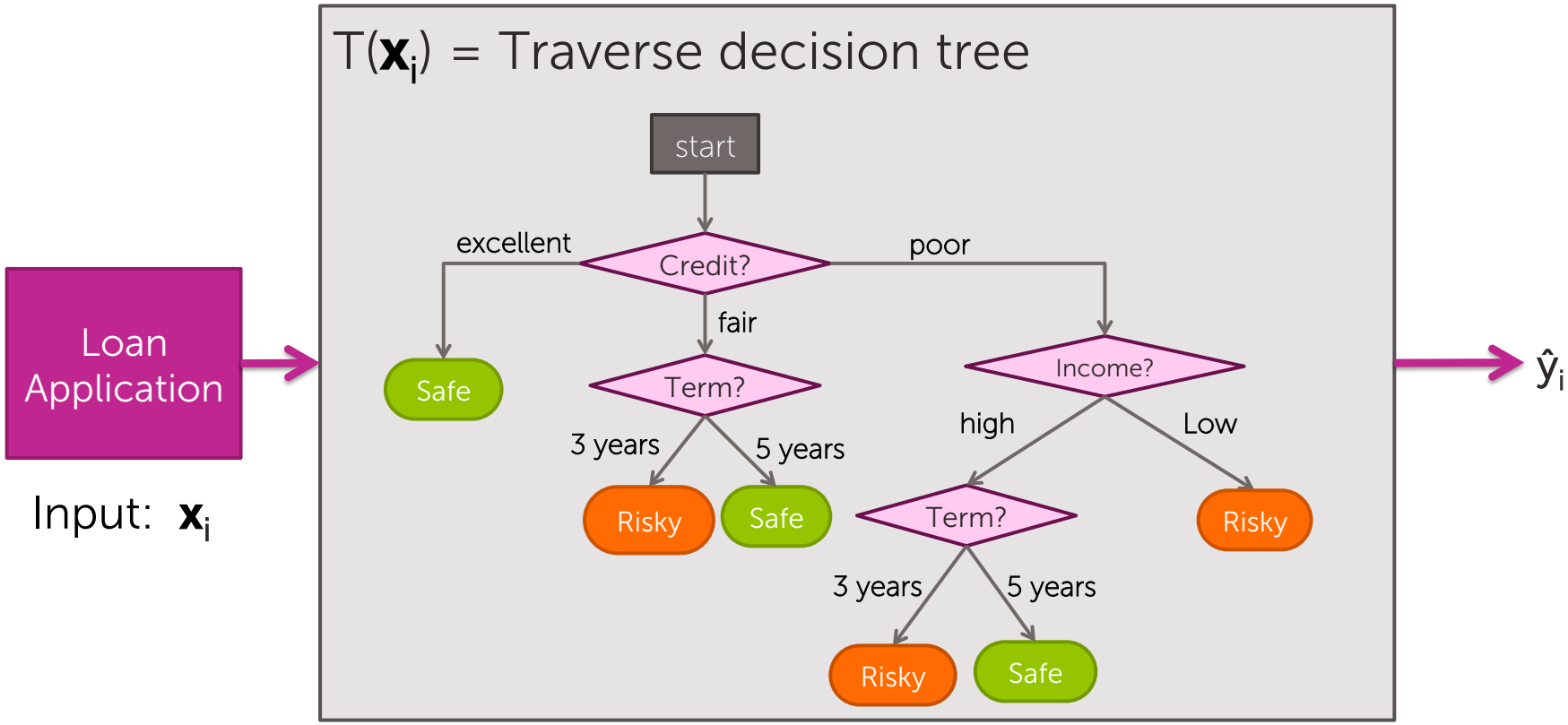
Handling missing data

©2015-2016 Emily Fox & Carlos Guestrin

Decision tree review



©2015-2016 Emily Fox & Carlos Guestrin

So far: data always completely observed

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	5 yrs	low	risky
fair	3 yrs	high	safe
poor	5 yrs	high	risky
excellent	3 yrs	low	risky
fair	5 yrs	low	safe
poor	3 yrs	high	risky
poor	5 yrs	low	safe
fair	3 yrs	high	safe

Known **x** and y values for all data points

©2015-2016 Emily Fox & Carlos Guestrin

Missing data

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	?	low	risky
fair	3 yrs	high	safe
poor	5 yrs	high	risky
excellent	3 yrs	low	risky
fair	5 yrs	high	safe
poor	?	high	risky
poor	5 yrs	low	safe
fair	?	high	safe

Loan application may be 3 or 5 years

©2015-2016 Emily Fox & Carlos Guestrin

Missing values impact training and predictions

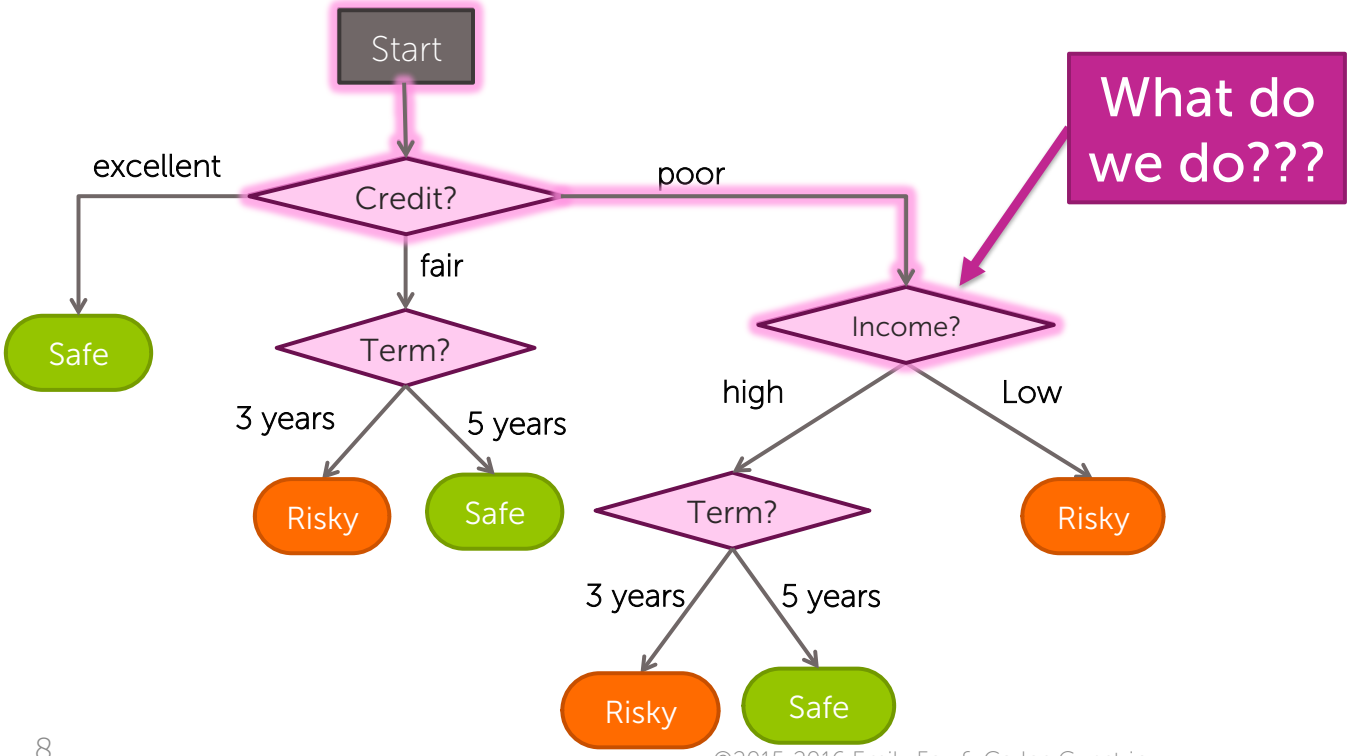
1. Training data: Contains “unknown” values

2. Predictions: Input at prediction time contains “unknown” values

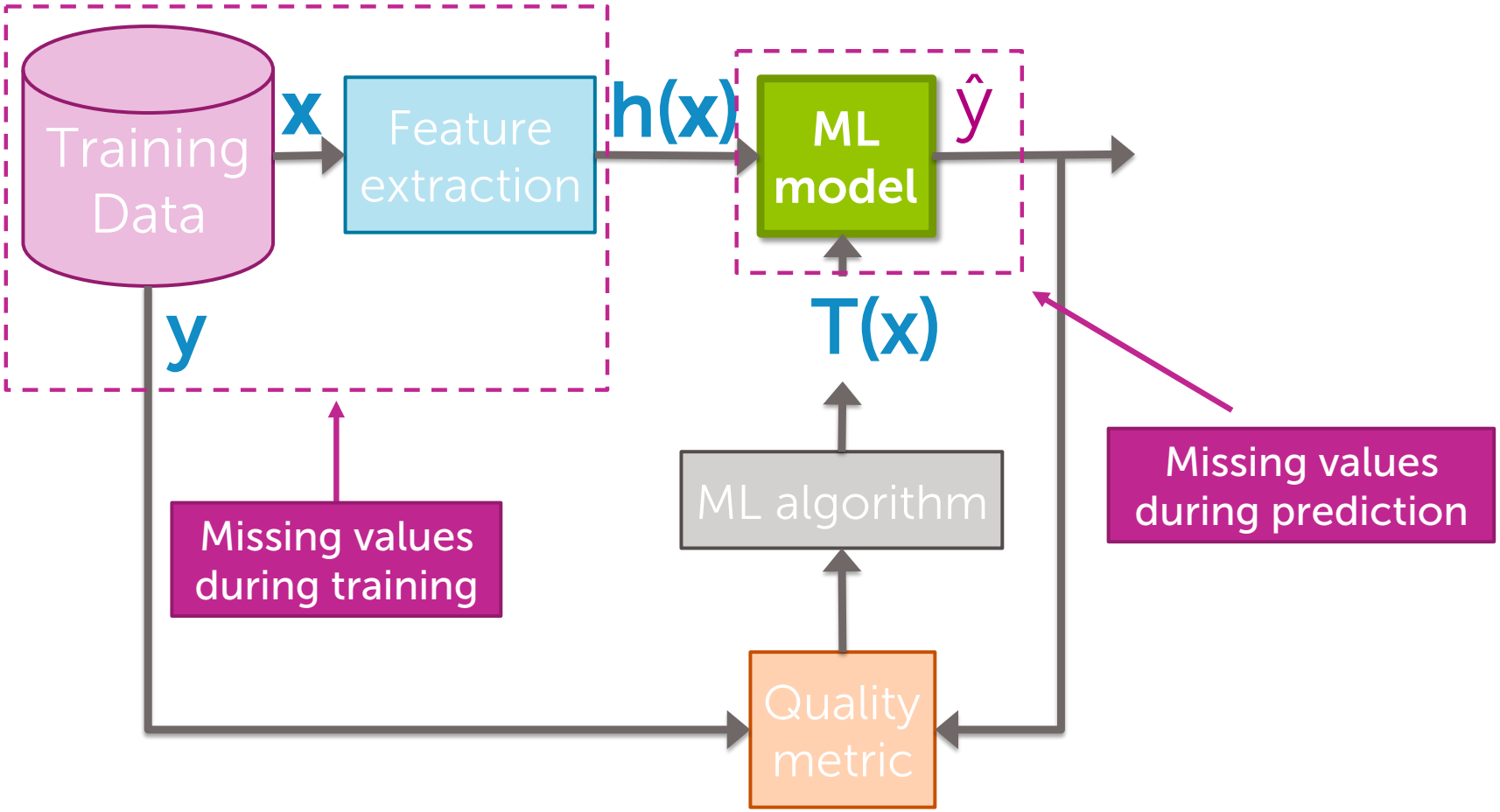
©2015-2016 Emily Fox & Carlos Guestrin

Missing values during prediction

$\mathbf{x}_i$ = (Credit = poor, Income = ?, Term = 5 years)



©2015-2016 Emily Fox & Carlos Guestrin



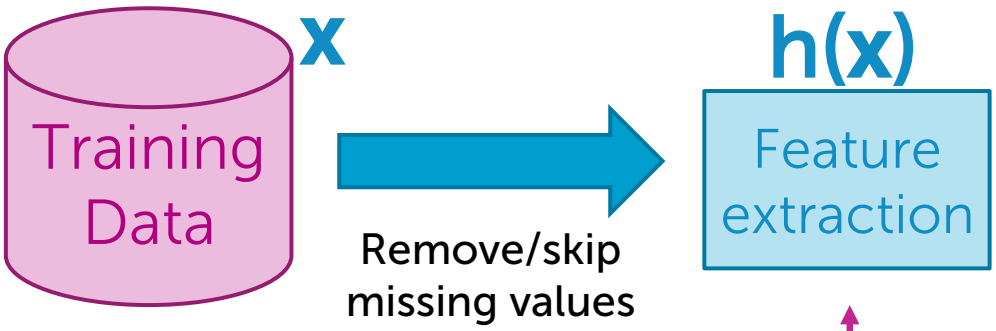
©2015-2016 Emily Fox & Carlos Guestrin

Handling missing data

Strategy 1: Purification by skipping

©2015-2016 Emily Fox & Carlos Guestrin

Idea 1: Purification by skipping/removing



May contain data points where one (or more) features are missing

h(**x**) is a subset of the data without any missing values

©2015-2016 Emily Fox & Carlos Guestrin

Idea 1: Skip data points with missing values

x

N = 9, 3 features

fewer

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	?	low	risky
fair	3 yrs	high	safe
poor	5 yrs	high	risky
excellent	3 yrs	low	risky
fair	5 yrs	high	safe
poor	3 yrs	low	risky
poor	3 yrs	?	safe
fair	?	high	safe

Skip data points with missing values

h(x)

N = 6, 3 features

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	3 yrs	high	safe
poor	5 yrs	high	risky
excellent	3 yrs	low	risky
fair	5 yrs	high	safe
poor	3 yrs	low	risky

©2015-2016 Emily Fox & Carlos Guestrin

The challenge with Idea 1

x

N = 9, 3 features

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	?	low	risky
fair	3 yrs	high	safe
poor	?	high	risky
excellent	?	low	risky
fair	?	high	safe
poor	3 yrs	low	risky
poor	?	low	safe
fair	?	high	safe

Warning: More than 50% of the loan terms are unknown!

much much smaller

Skip data points with missing values

h(x)

N = 3, 3 features

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	3 yrs	high	safe
poor	3 yrs	low	risky

©2015-2016 Emily Fox & Carlos Guestrin

Idea 2: Skip features with missing values

x

N = 9, 3 features

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	?	low	risky
fair	3 yrs	high	safe
poor	?	high	risky
excellent	?	low	risky
fair	5 yrs	high	safe
poor	?	high	risky
poor	?	low	safe
fair	?	high	safe

Skip features with many missing values

fewer features

h(x)

N = 9, 2 features

Credit	Income	y
excellent	high	safe
fair	low	risky
fair	high	safe
poor	high	risky
excellent	low	risky
fair	high	safe
poor	high	risky
poor	low	safe
fair	high	safe

©2015-2016 Emily Fox & Carlos Guestrin

Missing value skipping: Ideas 1 & 2

Idea 1: Skip data points where any feature contains a missing value

- Make sure only a few data points are skipped

Idea 2: Skip an entire feature if it's missing for many data points

- Make sure only a few features are skipped

©2015-2016 Emily Fox & Carlos Guestrin

Missing value skipping: Pros and Cons

Pros

- Easy to understand and implement
- Can be applied to any model (decision trees, logistic regression, linear regression,…)

Cons

- Removing data points and features may remove important information from data
- Unclear when it's better to remove data points versus features
- Doesn't help if data is missing at prediction time

©2015-2016 Emily Fox & Carlos Guestrin

Handling missing data

Strategy 2: Prification by imputing

©2015-2016 Emily Fox & Carlos Guestrin

Main drawback of skipping strategy

x

N = 9, 3 features

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	?	low	risky
fair	3 yrs	high	safe
poor	5 yrs	high	risky
excellent	3 yrs	low	risky
fair	5 yrs	high	safe
poor	3 yrs	low	risky
poor	?	low	safe
fair	?	high	safe

Data is precious, don't throw it away!

Skip data points with missing values

h(x)

N = 6, 3 features

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	3 yrs	high	safe
poor	5 yrs	high	risky
excellent	3 yrs	low	risky
fair	5 yrs	high	safe
poor	3 yrs	low	risky

©2015-2016 Emily Fox & Carlos Guestrin

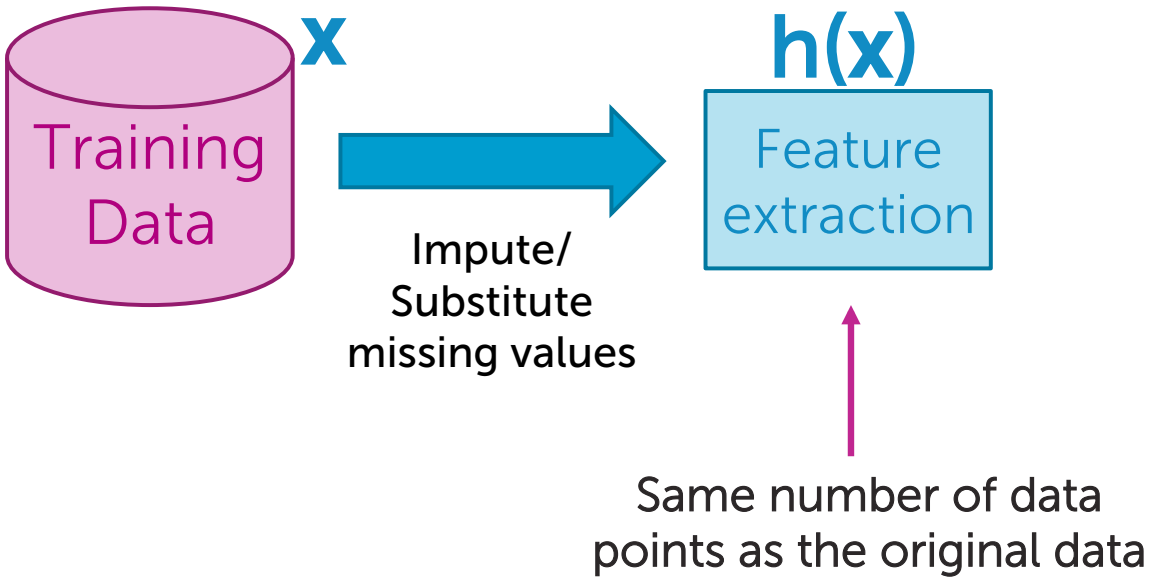
Can we keep all the data?

credit	term	income	y
excellent	3 yrs	high	safe
fair	?	low	risky
fair	3 yrs	high	safe
poor	5 yrs	high	risky
excellent	3 yrs	low	risky
fair	5 yrs	high	safe
poor	3 yrs	high	risky
poor	?	low	safe
fair	?	high	safe

Use other data points in **x** to "guess" the "?"

©2015-2016 Emily Fox & Carlos Guestrin

Idea 2: Purification by imputing



©2015-2016 Emily Fox & Carlos Guestrin

Idea 2: Imputation/Substitution

N = 9, 3 features

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	?	low	risky
fair	3 yrs	high	safe
poor	5 yrs	high	risky
excellent	3 yrs	low	risky
fair	5 yrs	high	safe
poor	3 yrs	high	risky
poor	?	low	safe
fair	?	high	safe

Fill in each missing value with a calculated guess

N = 9, 3 features

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	**3 yrs**	low	risky
fair	3 yrs	high	safe
poor	5 yrs	high	risky
excellent	3 yrs	low	risky
fair	5 yrs	high	safe
poor	3 yrs	high	risky
poor	**3 yrs**	low	safe
fair	**3 yrs**	high	safe

©2015-2016 Emily Fox & Carlos Guestrin

Example: Replace ? with most common value

3 year loans: 4
5 year loans: 2

Best simple guess

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	?	low	risky
fair	3 yrs	high	safe
poor	5 yrs	high	risky
excellent	3 yrs	low	risky
fair	5 yrs	high	safe
poor	3 yrs	high	risky
poor	?	low	safe
fair	?	high	safe

Purification by imputing

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	**3 yrs**	low	risky
fair	3 yrs	high	safe
poor	5 yrs	high	risky
excellent	3 yrs	low	risky
fair	5 yrs	high	safe
poor	3 yrs	high	risky
poor	**3 yrs**	low	safe
fair	**3 yrs**	high	safe

©2015-2016 Emily Fox & Carlos Guestrin

Common (simple) rules for purification by imputation

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	?	low	risky
fair	3 yrs	high	safe
poor	5 yrs	high	risky
excellent	3 yrs	low	risky
fair	5 yrs	high	safe
poor	3 yrs	high	risky
poor	?	low	safe
fair	?	high	safe

Impute each feature with missing values:

1. Categorical features use mode: Most popular value (mode) of non-missing $\mathbf{x}_i$
2. Numerical features use average or median: Average or median value of non-missing $\mathbf{x}_i$

Many advanced methods exist, e.g., expectation-maximization (EM) algorithm

©2015-2016 Emily Fox & Carlos Guestrin

Missing value imputation: Pros and Cons

Pros

- Easy to understand and implement
- Can be applied to any model
  (decision trees, logistic regression, linear regression,…)
- Can be used at prediction time: use same imputation rules

Cons

- May result in systematic errors

Example: Feature "age" missing in all banks in Washington by state law

©2015-2016 Emily Fox & Carlos Guestrin

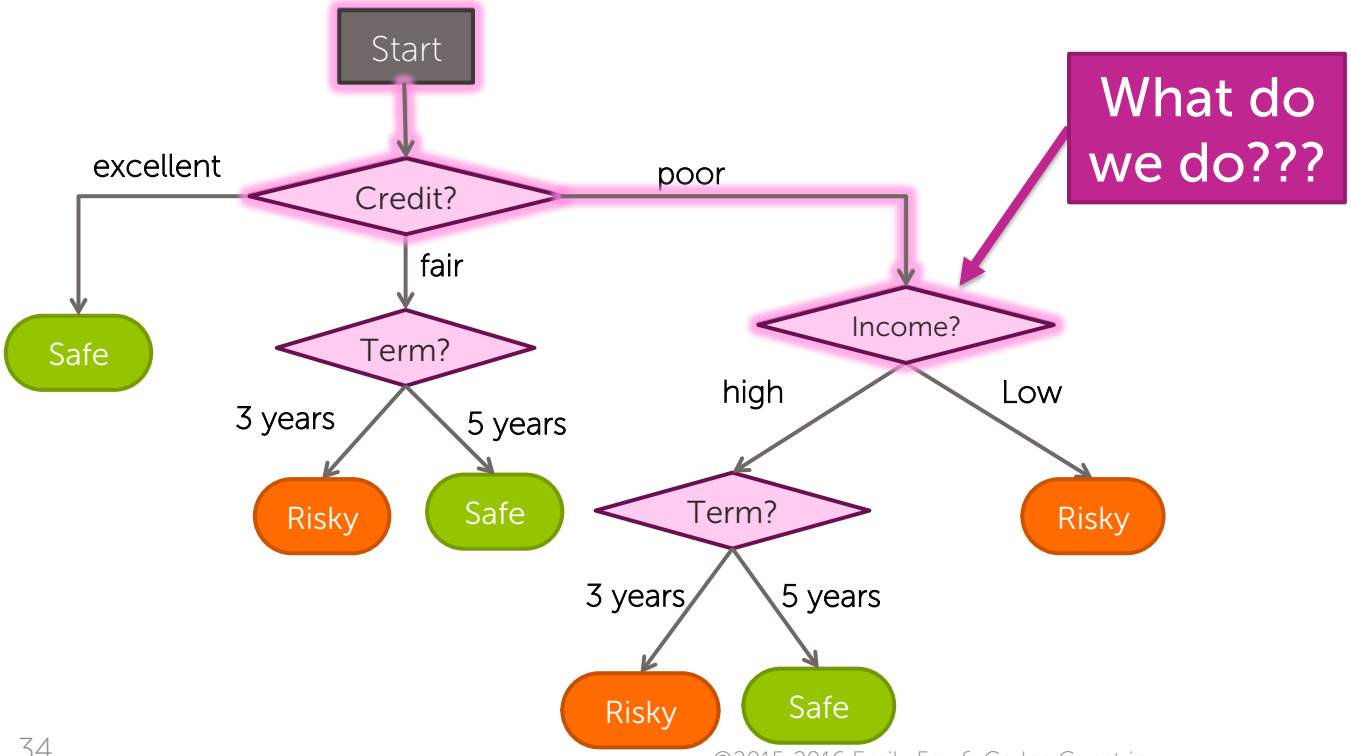
Handling missing data

***Strategy 3**: Adapt learning algorithm to be robust to missing values*

©2015-2016 Emily Fox & Carlos Guestrin

Missing values during prediction: *revisited*

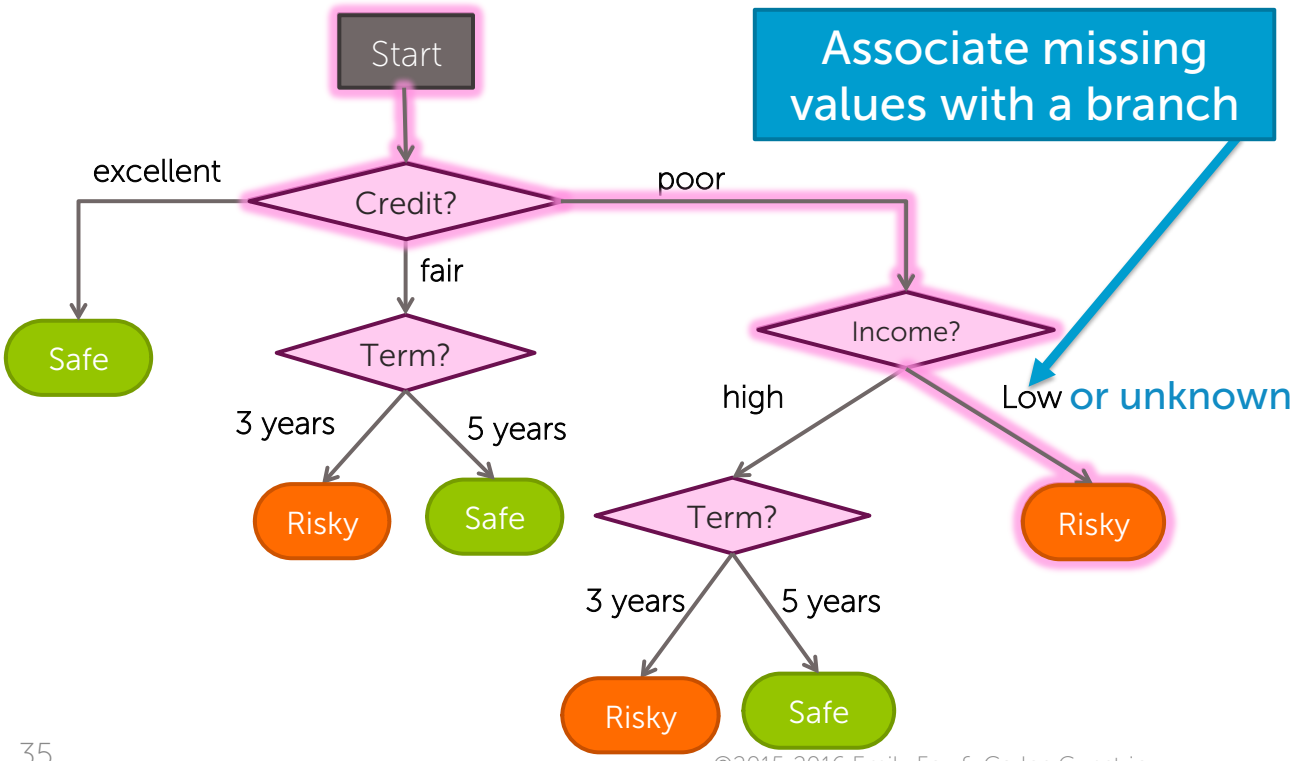
$\mathbf{x}_i$ = (Credit = poor, Income = ?, Term = 5 years)



©2015-2016 Emily Fox & Carlos Guestrin

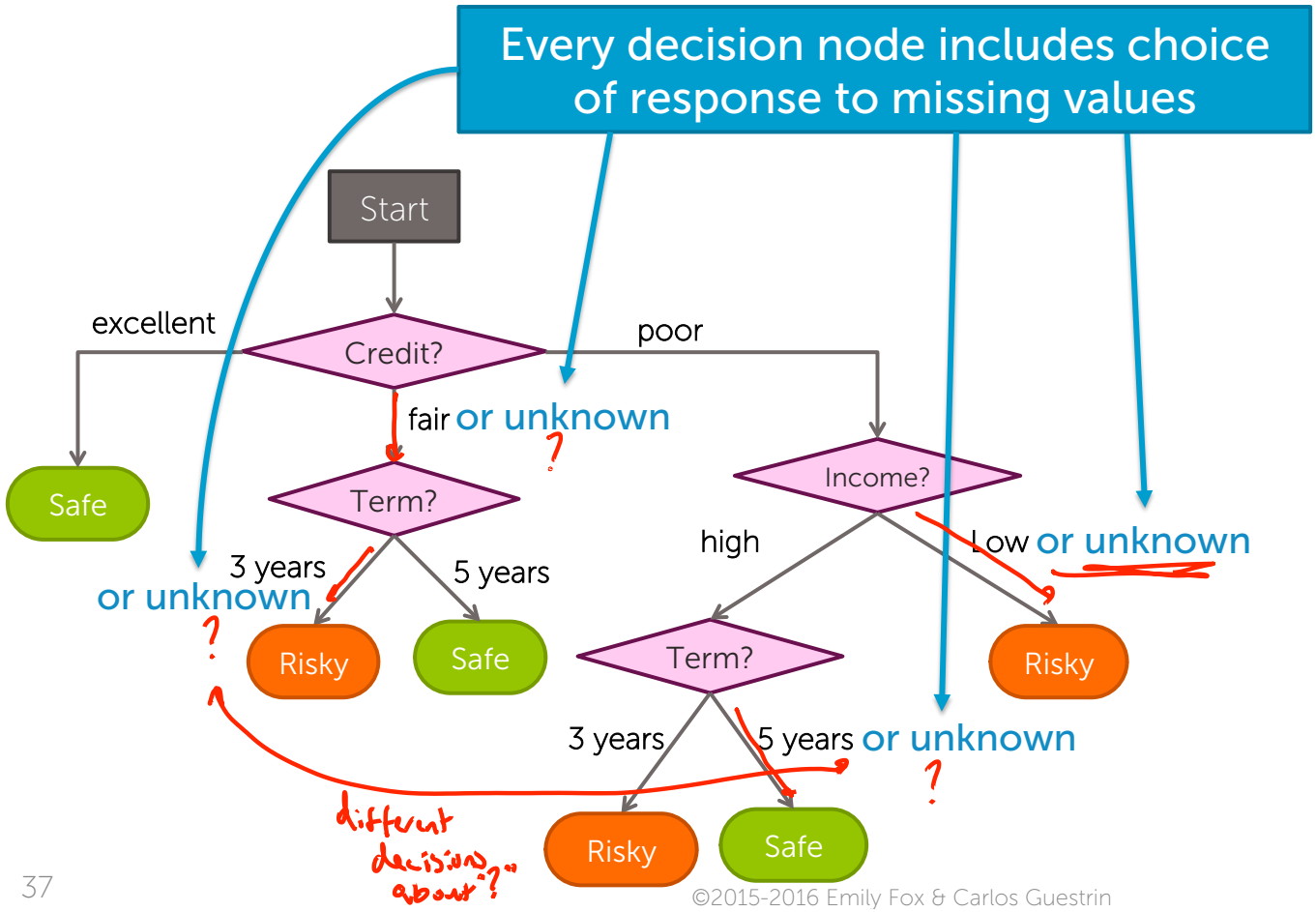
Add missing values to the tree definition

$\mathbf{x}_i$ = (Credit = poor, Income = ?, Term = 5 years)



©2015-2016 Emily Fox & Carlos Guestrin

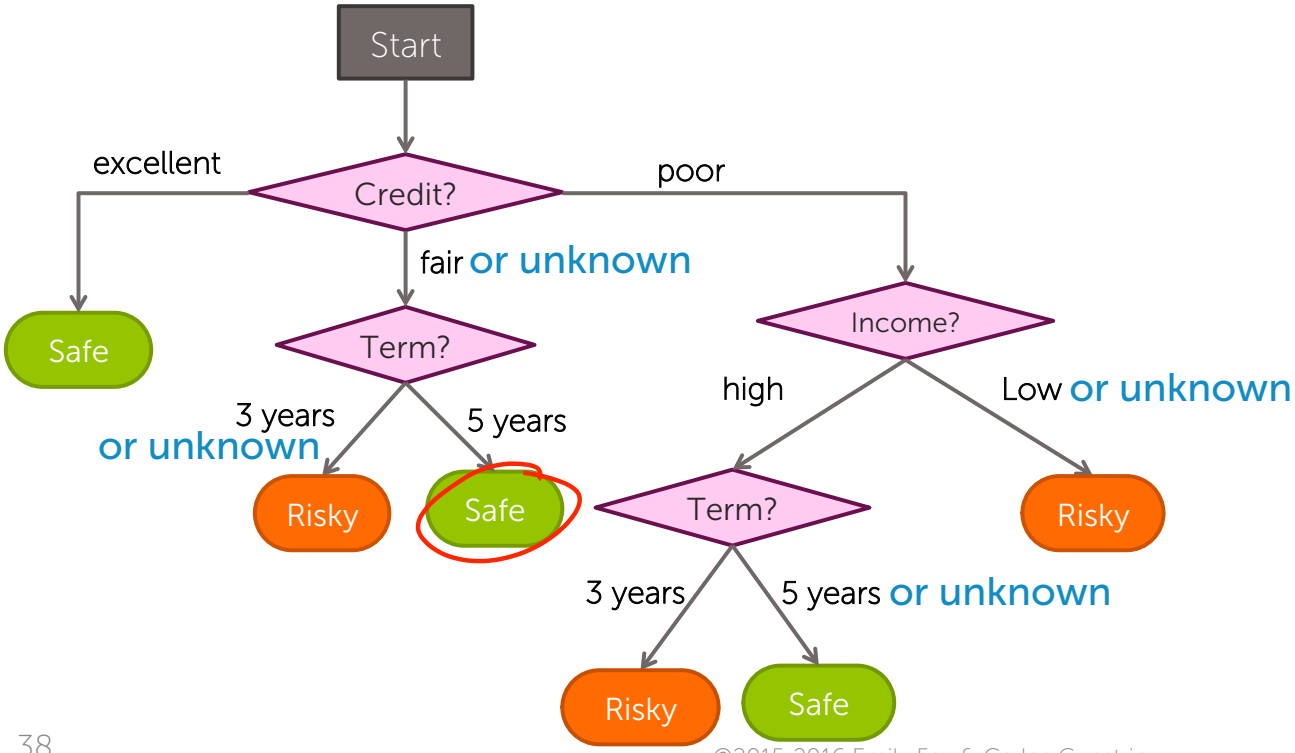
Add missing value choice to every decision node



©2015-2016 Emily Fox & Carlos Guestrin

Prediction with missing values becomes simple

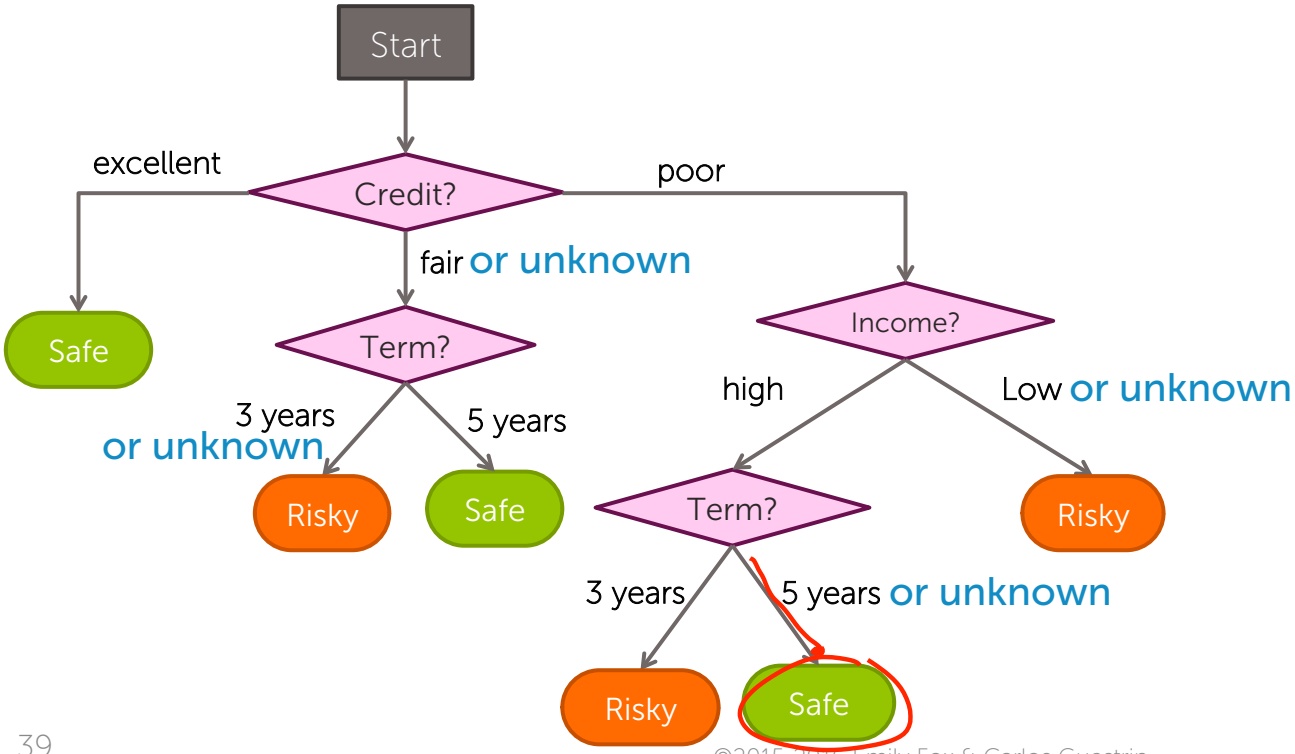
$\mathbf{x}_i$ = (Credit = ?, Income = high, Term = 5 years)



©2015-2016 Emily Fox & Carlos Guestrin

Prediction with missing values becomes simple

$\mathbf{x}_i$ = (Credit = poor, Income = high, Term = ?)



©2015-2016 Emily Fox & Carlos Guestrin

Explicitly handling missing data by learning algorithm: Pros and Cons

Pros

- Addresses training and prediction time
- **More accurate predictions**

Cons

- Requires modification of learning algorithm
  - Very simple for decision trees

©2015-2016 Emily Fox & Carlos Guestrin

Feature split selection with missing data

©2015-2016 Emily Fox & Carlos Guestrin

Greedy decision tree learning

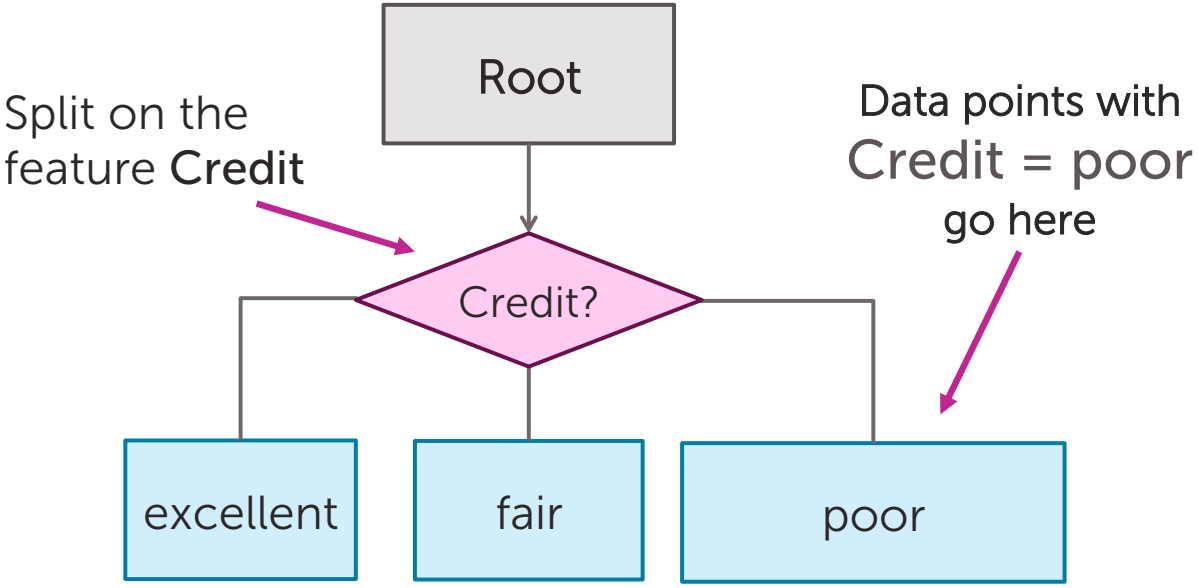
- Step 1: Start with an empty tree
- Step 2: Select a feature to split data
- For each split of the tree:
  - Step 3: If nothing more to, make predictions
  - Step 4: Otherwise, go to Step 2 & continue (recurse) on this split

Pick feature split leading to lowest classification error

Must select feature & branch for missing values!

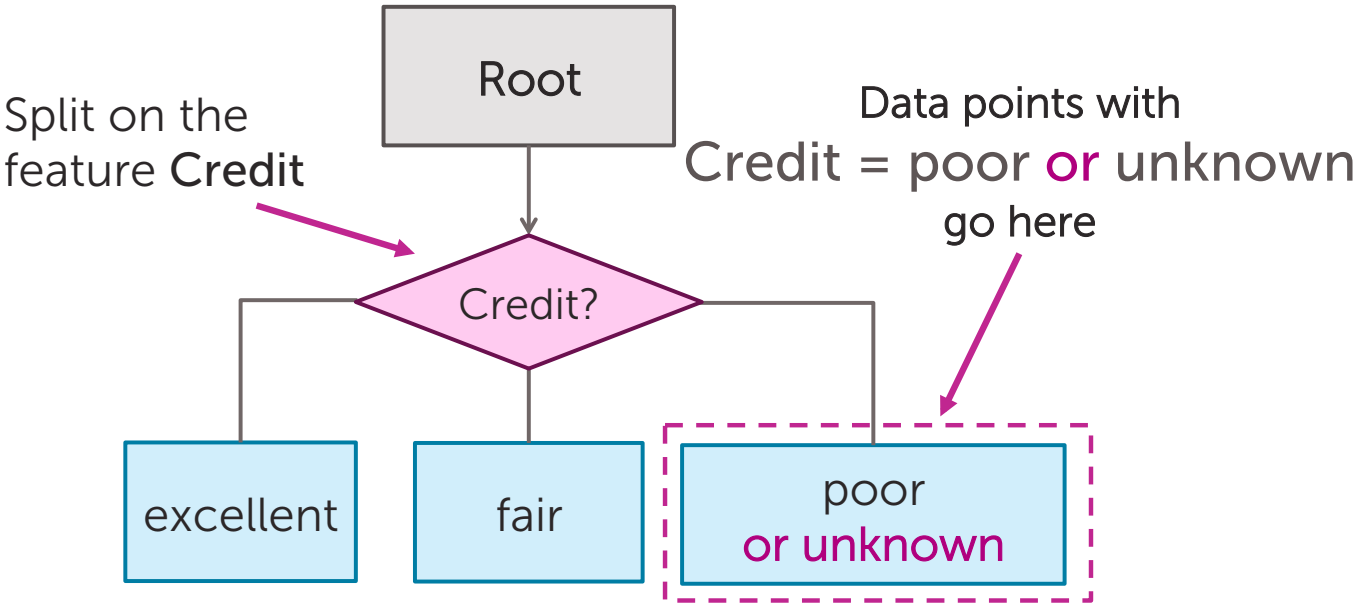
©2015-2016 Emily Fox & Carlos Guestrin

Feature split (without missing values)



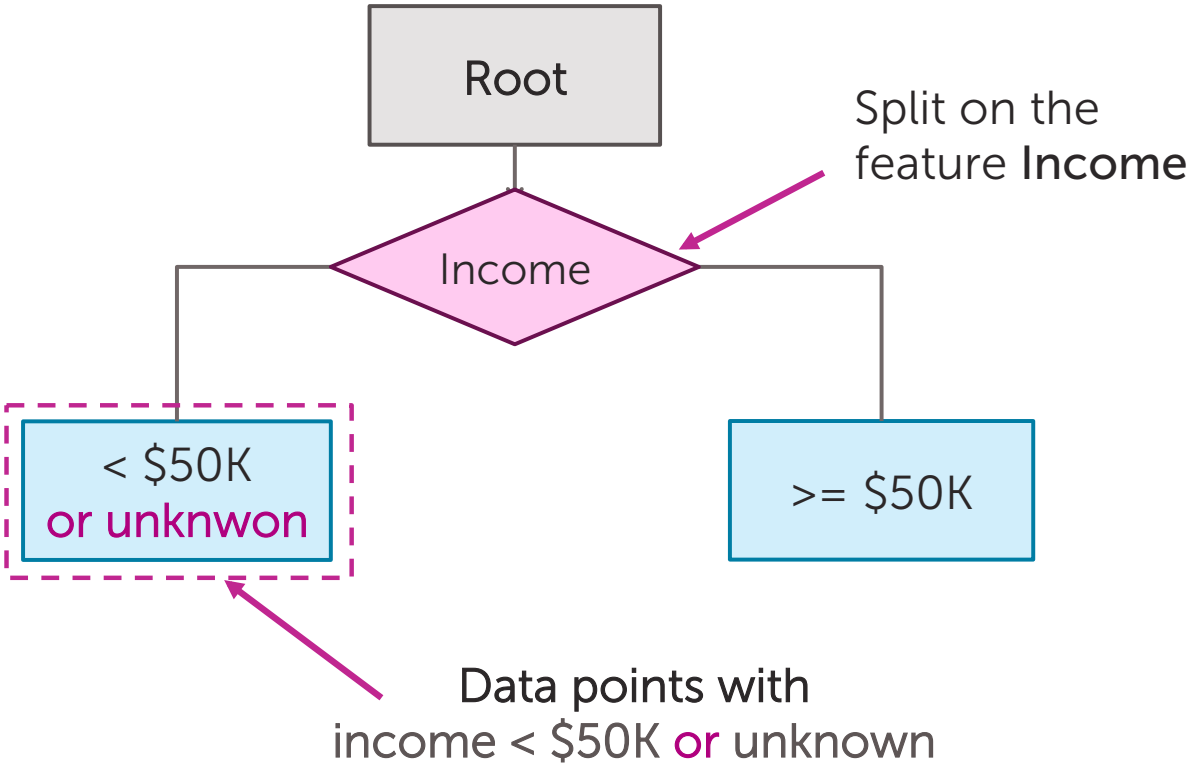
©2015-2016 Emily Fox & Carlos Guestrin

Feature split (with missing values)



©2015-2016 Emily Fox & Carlos Guestrin

Missing value handling in threshold splits

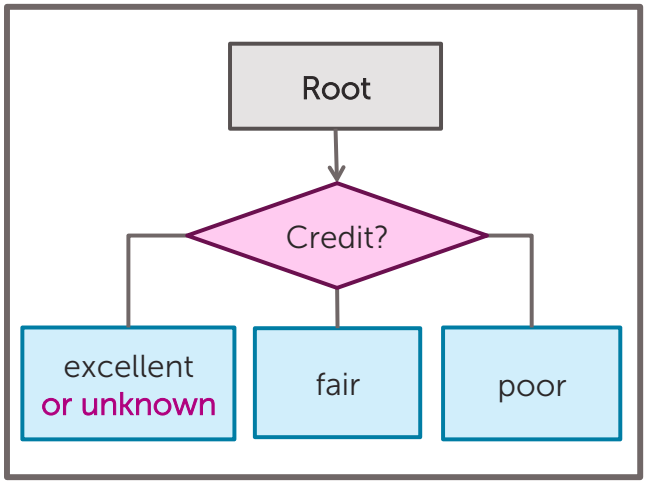


©2015-2016 Emily Fox & Carlos Guestrin

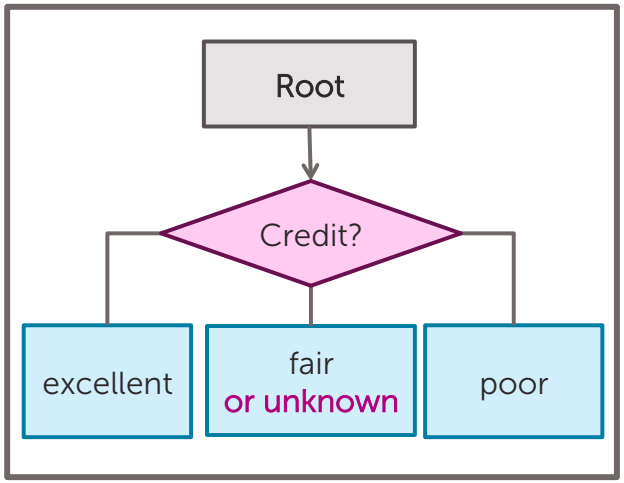
Should missing go left, right, or middle?

Choose branch that leads to lowest classification error!

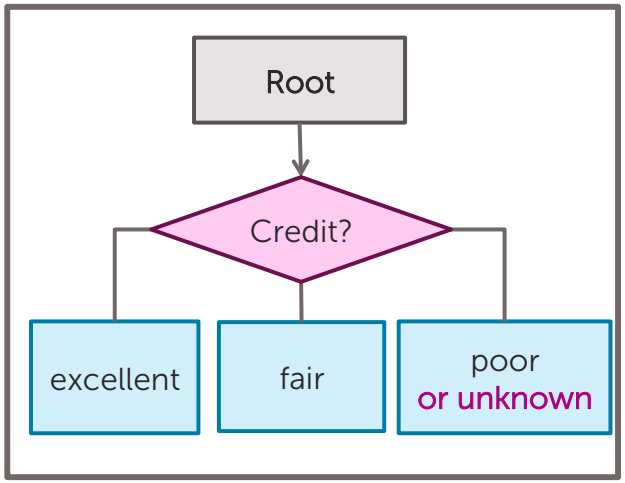
Choice 1: Missing values go with Credit=excellent



Choice 2: Missing values go with Credit=fair



Choice 3: Missing values go with Credit=poor



©2015-2016 Emily Fox & Carlos Guestrin

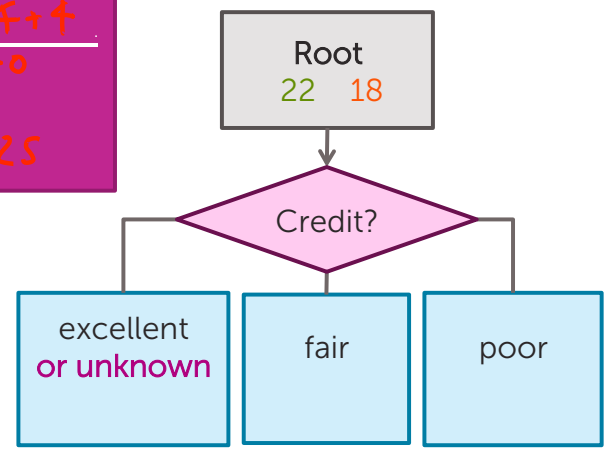
Computing classification error of decision stump with missing data

N = 40, 3 features

Credit	Term	Income	y
excellent	3 yrs	high	safe
?	5 yrs	low	risky
fair	3 yrs	high	safe
poor	5 yrs	high	risky
?	3 yrs	low	risky
?	5 yrs	low	safe
poor	3 yrs	high	risky
poor	5 yrs	low	safe
fair	3 yrs	high	safe
…	…	…	…

Error = $\frac{2+4+4}{40}$

= 0.25



Observed values	9 0	8 4	4 12

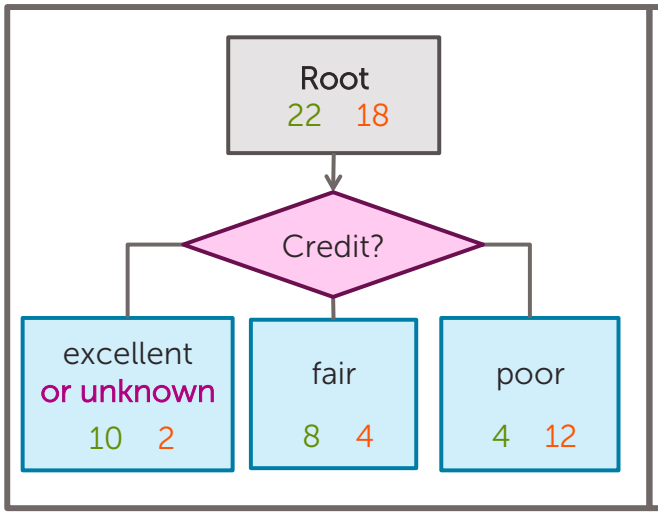
0 0 0

©2015-2016 Emily Fox & Carlos Guestrin

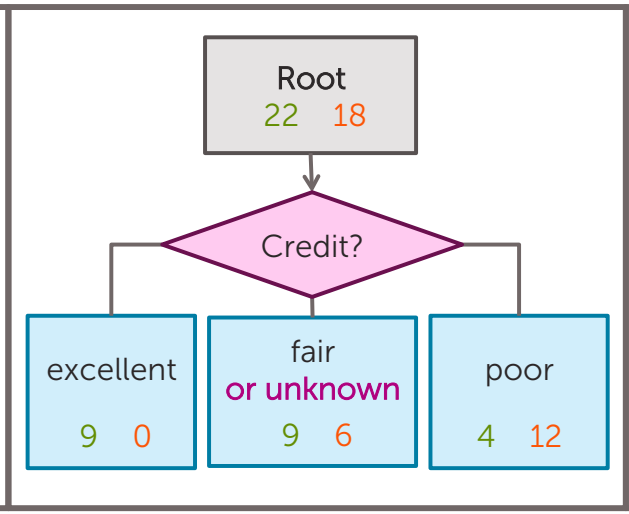
Use classification error to decide

Best choice → assign "unknown" to Credit = poor

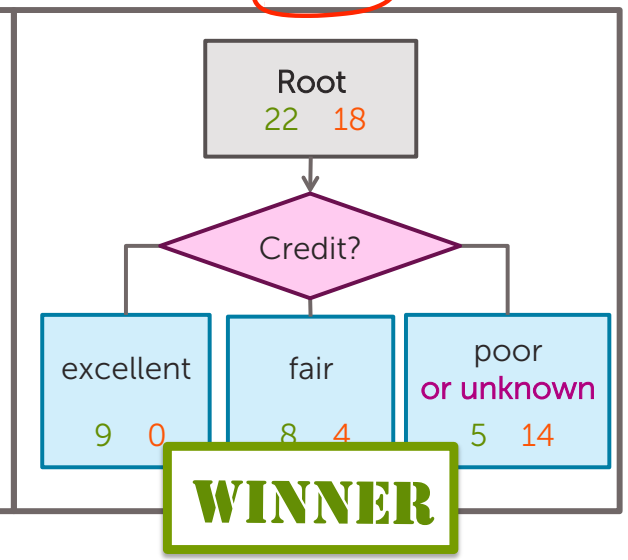
Choice 1: error = 0.25



Choice 2: error = 0.25



Choice 3: error = 0.225



©2015-2016 Emily Fox & Carlos Guestrin

Feature split selection algorithm with missing value handling

- Given a subset of data M (a node in a tree)
- For each feature $h_i(x)$:
  1. Split data points of M where $h_i(x)$ is *not* "unknown" according to feature $h_i(x)$
  2. Consider assigning data points with "unknown" value for $h_i(x)$ to each branch
     A. Compute classification error split & branch assignment of "unknown" values
- Chose feature $h^*(x)$ & branch assignment of "unknown" with lowest classification error

©2015-2016 Emily Fox & Carlos Guestrin

Summary of handling missing data

©2015-2016 Emily Fox & Carlos Guestrin

What you can do now…

Describe common ways to handling missing data:

1. Skip all rows with any missing values
2. Skip features with many missing values
3. Impute missing values using other data points

Modify learning algorithm (decision trees) to handle missing data:

1. Missing values get added to one branch of split
2. Use classification error to determine where missing values go

©2015-2016 Emily Fox & Carlos Guestrin

Thank you to Dr. Krishna Sridhar

Dr. Krishna Sridhar

Staff Data Scientist, Dato, Inc.

©2015-2016 Emily Fox & Carlos Guestrin